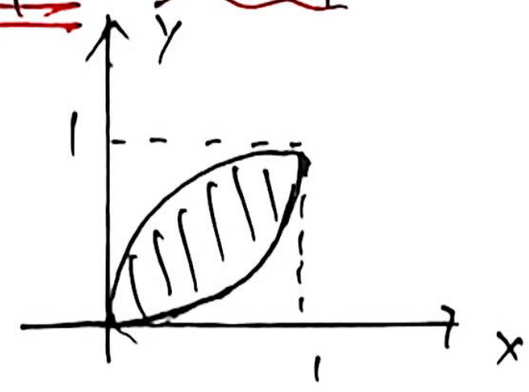
2026.03.10. 重积分.

一、累次积分方法.

1. $\iint_D (x^2+2y)\,dxdy$. $D: y=x^2,\ y=\sqrt{x}$ 所围.

Step1. 画图了解积分区域 D.



$\xrightarrow{\text{累次化}}$ $\int_0^1\int_{x^2}^{\sqrt{x}} ?\,dy\,dx$

$\int_0^1\int_{y^2}^{\sqrt{y}} ?\,dx\,dy$.

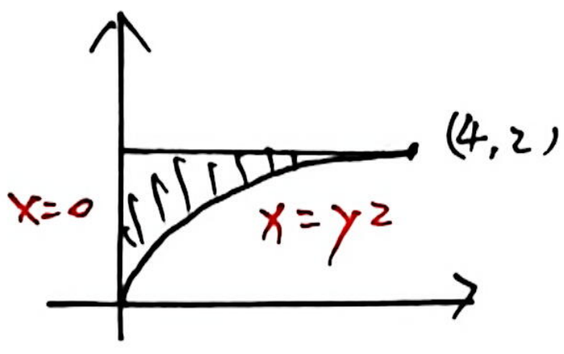
Step2. 对不同被积函数，选择合适方法.

$$原式 = \int_0^1\int_{x^2}^{\sqrt{x}} x^2\,dy\,dx + \int_0^1\int_{y^2}^{\sqrt{y}} 2y\,dx\,dy$$

$$= \int_0^1 x^{5/2}-x^4\,dx + 2\int_0^1 y^{3/2}-y^3\,dy$$

$$= \frac{2}{7}-\frac{1}{5}+2\left(\frac{2}{5}-\frac{1}{4}\right) = \frac{27}{70}.$$

2. $\iint_D \sin y^3\,dxdy$. $D: y=\sqrt{x},\ y=2,\ x=0$.



由于 $\int \sin y^3\,dy$ 难做，故先积 x.

$$原式 = \int_0^2\int_0^{y^2} \sin y^3\,dxdy = \int_0^2 y^2\sin y^3\,dy$$

$$= \frac{1}{3}\int_0^2 \sin y^3\,dy^3 = \frac{1}{3}\int_0^8 \sin t\,dt \quad (t=y^3)$$

$$= \frac{1}{3}(1-\cos 8).$$

3. $\iiint_D x^2y^4\sin z\,dxdydz$. $D: x^2+y^2+z^2\le 1$

因为 D 关于 xy-平面对称，$x^2y^4\sin z$ 是 z 的奇函数

原式 = 0.

4. $\int\cdots\int_D \left(\sum_{i=1}^{n} x_i\right)^2 dx_1\cdots dx_n \qquad D=[0,1]^n.$

$$= \int\cdots\int_D \sum_{i=1}^{n} x_i^2 + 2\sum_{1\le i<j\le n} x_i x_j \, dx_1\cdots dx_n.$$

① $\int\cdots\int_D x_i^2\, dx_1\cdots dx_n = \int_0^1 x_i^2\, dx_i = \frac{1}{3}.$

② $\int\cdots\int_D x_i x_j\, dx_1\cdots dx_n = \int_0^1\int_0^1 x_i x_j\, dx_i\, dx_j = \frac{1}{4}$

$\Rightarrow$ 原式 $= \frac{n}{3} + 2\times\frac{n(n-1)}{2}\times\frac{1}{4} = \frac{n^2}{4}+\frac{n}{12}.$

二、极换元.

当出现 x^2+y^2. 常考虑极坐标 $\begin{cases} x = r\cos\theta \\ y = r\sin\theta. \end{cases}$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = r.$$

一般形式为椭圆参数方程. $\frac{(x-x_0)^2}{a^2}+\frac{(y-y_0)^2}{b^2}\le 1 \rightsquigarrow \boxed{\begin{cases} x = x_0 + ar\cos\theta \\ y = y_0 + br\sin\theta \end{cases}}.$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = abr.$$

1. $\iint_D (x^2+x^4y)\,dx\,dy. \qquad D: 1\le x^2+y^2\le 4$

即 $1\le r\le 2$. 提示极坐标.

$$= \iint_D (r^2\cos^2\theta + \underbrace{r^5\cos^4\theta\sin\theta}_{\text{积分}=0})\cdot r\,dr\,d\theta.$$

$$= \underbrace{\int_1^2 r^3\,dr}_{=\frac{15}{4}} \underbrace{\int_0^{2\pi}\cos^2\theta\,d\theta}_{=\int_0^{2\pi}\frac{1}{2}(1+\cos 2\theta)\,d\theta = \pi}$$

$$= \frac{15}{4}\pi.$$

2. $\iint_D xy\,dx\,dy$. $D: \underbrace{\frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1}_{0\le r\le 1}$ 在 $\underbrace{\text{I象限部分}}_{0\le\theta\le\frac{\pi}{2}}$.

$$\begin{cases} x = ar\cos\theta \\ y = br\sin\theta. \end{cases}$$

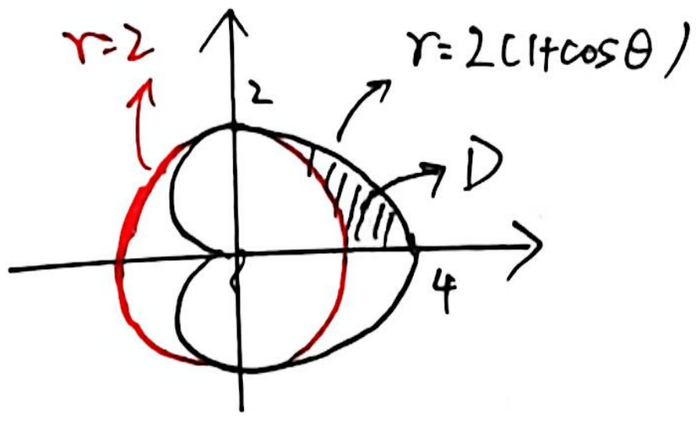
原式 $= \iint_D abr^2\cos\theta\sin\theta\cdot abr\,dr\,d\theta$

$$= a^2b^2\underbrace{\int_0^1 r^3\,dr}_{=\frac{1}{4}}\underbrace{\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\,d\theta}_{=\frac{1}{2}\int_0^{\frac{\pi}{2}}\sin 2\theta\,d\theta \;=\; \frac{1}{4}\int_0^{\pi}\sin t\,dt \;=\; \frac{1}{2}}$$

$t=2\theta$

$$= \frac{a^2b^2}{8}.$$

3. $\iint_D y\,dx\,dy$. $D: \underbrace{r\le 2(1+\cos\theta),\ r\ge 2}_{r\text{上下限与}\theta\text{有关，先积}r}$. $\underbrace{\text{I象限}}_{0\le\theta\le\frac{\pi}{2}}$.



原式 $= \int_0^{\frac{\pi}{2}}\int_2^{2+2\cos\theta} r\sin\theta\cdot r\,dr\,d\theta$

$$= \int_0^{\frac{\pi}{2}}\sin\theta\cdot\frac{1}{3}r^3\Big|_2^{2+2\cos\theta}d\theta$$

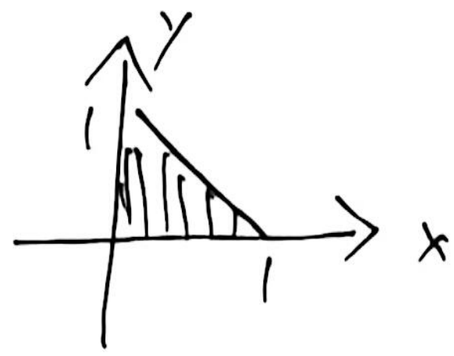
$$= \frac{8}{3}\int_0^{\frac{\pi}{2}}\sin\theta\left(\cos^3\theta+3\cos^2\theta+3\cos\theta\right)d\theta$$

$$= \frac{8}{3}\int_0^1(c^3+3c^2+3c)\,dc. \quad (c=\cos\theta)$$

$$= \frac{8}{3}\left(\frac{1}{4}+3\times\frac{1}{3}+3\times\frac{1}{2}\right) = \frac{22}{3}.$$

三、一般换元.

1. $\iint_D e^{\frac{y}{x+y}}\,dx\,dy$. $D: x=0,\ y=0,\ x+y=1$



无法积分之困难在于分母上的 $x+y$. 故对其换元.

$t=x+y$. $\frac{\partial(t,y)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial t}{\partial x} & \frac{\partial t}{\partial y} \\ \frac{\partial y}{\partial x} & \frac{\partial y}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1.$

$$\Rightarrow \boxed{\frac{\partial(x,y)}{\partial(t,y)} = \left(\frac{\partial(t,y)}{\partial(x,y)}\right)^{-1} = 1}$$

☆注意此关系. 对于复杂非线性换元, 常用此关系算Jacobian.

积分区域 t 从 0 到 1, y 从 0 到 t.

$$\Rightarrow \text{原式} = \int_0^1\int_0^t e^{\frac{y}{t}}\cdot 1\,dy\,dt = \int_0^1 te^{\frac{y}{t}}\Big|_{y=0}^{y=t}dt = (e-1)\int_0^1 t\,dt$$

$$= \frac{e-1}{2}.$$

2. 求 $\left(\frac{x}{\sqrt{2}}+\frac{y}{\sqrt{3}}\right)^2+\frac{z^2}{2}=1$ 在第I卦限部分体积

Step1. 先找积分区域，找该几何体与 xy, yz, zx 三个平面之交线.

$x=y=0 \Rightarrow z=\sqrt{2}$

$x=z=0 \Rightarrow y=\sqrt{3}$

$y=z=0 \Rightarrow x=\sqrt{2}$

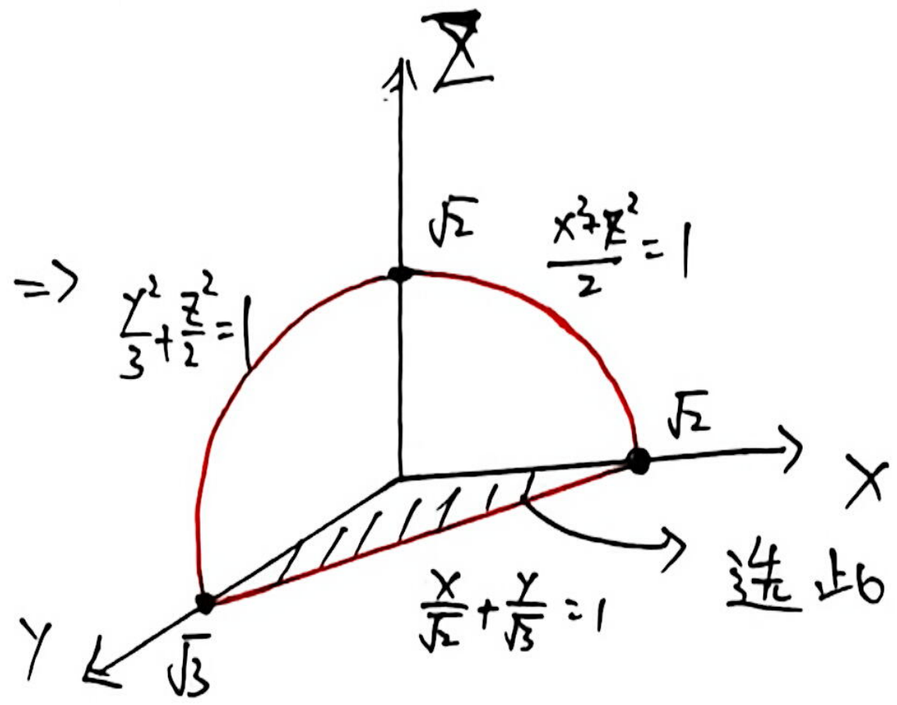
与轴交点

$z=0 \Rightarrow \frac{x}{\sqrt{2}}+\frac{y}{\sqrt{3}}=1$

$x=0 \Rightarrow \frac{y^2}{3}+\frac{z^2}{2}=1$

$y=0 \Rightarrow \frac{x^2}{2}+\frac{z^2}{2}=1$

与面交线.



选此三角形为 D. $\begin{cases} x\geq 0 \\ y\geq 0 \\ \frac{x}{\sqrt{2}}+\frac{y}{\sqrt{3}}\leq 1. \end{cases}$

Step2. 确定被积函数.

x, y 固定后，$0\leq z\leq \sqrt{2}\cdot\sqrt{1-\left(\frac{x}{\sqrt{2}}+\frac{y}{\sqrt{3}}\right)^2}$.

因此. $f(x,y)=\sqrt{2}\cdot\sqrt{1-\left(\frac{x}{\sqrt{2}}+\frac{y}{\sqrt{3}}\right)^2}$.

Step3. 积分.

$$V=\sqrt{2}\cdot\iint_D\sqrt{1-\left(\frac{x}{\sqrt{2}}+\frac{y}{\sqrt{3}}\right)^2}\,dx\,dy.$$

换元 $\begin{cases} u=\frac{x}{\sqrt{2}}+\frac{y}{\sqrt{3}} \\ v=\frac{x}{\sqrt{2}}-\frac{y}{\sqrt{3}} \end{cases}$

$$\frac{\partial(u,v)}{\partial(x,y)}=\begin{vmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \end{vmatrix}=-\frac{2}{\sqrt{6}}$$

$$\Rightarrow \left|\frac{\partial(x,y)}{\partial(u,v)}\right|=\frac{\sqrt{6}}{2}$$

Rmk. 这是线性换元的好处. Jacobian 是个常数！因此除非特殊结构，优先考虑线性换元.

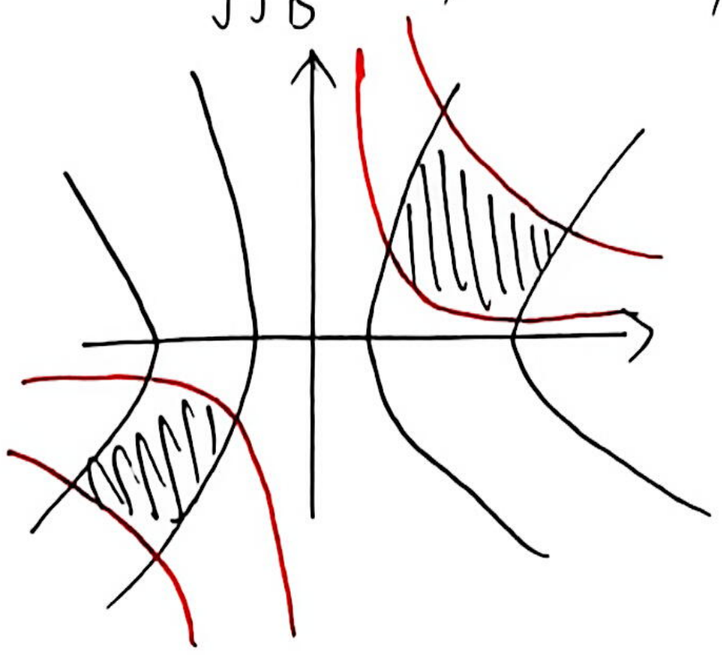
积分上下限：u 从 0 到 1.

$\left.\begin{array}{l} u+v=\sqrt{2}x\geq 0 \\ u-v=\frac{2}{\sqrt{3}}y\geq 0 \end{array}\right\}$ v 从 $-u$ 到 u

$$\Rightarrow V=\sqrt{2}\cdot\int_0^1\int_{-u}^{u}\sqrt{1-u^2}\cdot\frac{\sqrt{6}}{2}\,dv\,du=\sqrt{3}\int_0^1\sqrt{1-u^2}\cdot 2u\,du$$

$$=\sqrt{3}\int_0^1\sqrt{1-t}\,dt \quad (t=u^2) \quad =\frac{2}{3}\sqrt{3}.$$

3. $\iint_D (x^2+y^2)\,dxdy$. $D: 1 \le x^2-y^2 \le 9,\ 2 \le xy \le 4$.



D有2块. 原因是 $\begin{cases} u = x^2-y^2 \\ v = xy \end{cases}$ 不是双射

$(u,v) \longmapsto (x,y)$ 是二值映射, 需约束 (x,y) 在 I 还是 III 象限, 才确定值.

Jacobian $\dfrac{\partial(x,y)}{\partial(u,v)}$ 是对双射而言的, 因此需在 I 象限积分, 再 ×2.

I象限部分 $= \iint_D (x^2+y^2)\left|\dfrac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$. --- (△)

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 2x & -2y \\ y & x \end{vmatrix} = 2(x^2+y^2)$$

$$\Rightarrow \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \frac{1}{2(x^2+y^2)}$$

$$\Rightarrow (\triangle) = \iint_D \frac{1}{2}\,du\,dv = \frac{1}{2}\int_2^4\int_1^9 du\,dv = \frac{1}{2}\times 2\times 8 = 8.$$

$\Rightarrow$ 原积分 $= 16$.

四、一个柱坐标的例子. (匀质球对球外一点, 之万有引力).

$c>1$, 计算 $\displaystyle\int_{x^2+y^2+z^2\le 1} \frac{c-x}{((x-c)^2+y^2+z^2)^{3/2}}\,dx\,dy\,dz$.

① 准备工作.

$$\int_{-1}^{1}\frac{dx}{\sqrt{a-bx}} = -\frac{2}{b}\sqrt{a-bx}\Big|_{-1}^{1} = \frac{2}{b}\left(\sqrt{a+b}-\sqrt{a-b}\right). \quad \text{--- (i)}$$

$$\int_{-1}^{1}\frac{x\,dx}{\sqrt{a-bx}} = \frac{1}{b}\left(\int_{-1}^{1}\frac{a}{\sqrt{a-bx}}\,dx - \int_{-1}^{1}\sqrt{a-bx}\,dx\right)$$

$$= \frac{a}{b}\int_{-1}^{1}\frac{dx}{\sqrt{a-bx}} + \frac{2}{3b^2}(a-bx)^{3/2}\Big|_{-1}^{1}$$

$$= \frac{2a}{b^2}\left(\sqrt{a+b}-\sqrt{a-b}\right) - \frac{2}{3b^2}\left((a+b)^{3/2}-(a-b)^{3/2}\right). \quad \text{--- (ii)}$$

② 为什么用柱坐标.

被积函数中 $(x-c)^2+y^2+z^2$ 与积分区域 $x^2+y^2+z^2\le 1$

2个球心不匹配！球坐标可能很复杂

但两者皆关于 x 轴旋转对称. 故适合柱换元.

$$\begin{cases} y = r\cos\theta & 0\le r\le\sqrt{1-x^2} \\ z = r\sin\theta. & 0\le\theta\le 2\pi. \end{cases}$$

这一步 Jacobian 为 r.

③ 积分.

$$\text{原式} = \int_{x^2+y^2+z^2\le 1} \frac{c-x}{((x-c)^2+r^2)^{3/2}}\cdot r\, dr\, d\theta\, dx.$$

$$\int_0^{2\pi} d\theta \int_{-1}^{1} dx \int_0^{\sqrt{1-x^2}} dr$$

整体与 θ 无关.

$$= 2\pi \int_{-1}^{1}\int_0^{\sqrt{1-x^2}} \frac{c-x}{((x-c)^2+r^2)^{3/2}}\cdot r\, dr\, dx$$

$$= \pi \int_{-1}^{1}\int_0^{\sqrt{1-x^2}} \underbrace{\frac{c-x}{((x-c)^2+r^2)^{3/2}}\, dr^2}\, dx$$

关于 $t=r^2$ 积一个形如 $(\alpha+t)^{-\frac{3}{2}}$

原函数是 $-2(\alpha+t)^{-\frac{1}{2}}$.

$$= \pi \int_{-1}^{1} (-2)\cdot \overset{(c-x)}{\gamma}((x-c)^2+r^2)^{-\frac{1}{2}}\Big|_{r=0}^{r=\sqrt{1-x^2}}\, dx$$

$$= 2\pi \int_{-1}^{1} (c-x)\cdot\left(\frac{1}{\underbrace{|x-c|}_{c>1\ge x}} - \frac{1}{\sqrt{(x-c)^2+1-x^2}}\right) dx$$

$$= 2\pi\int_{-1}^{1} dx - 2\pi\int_{-1}^{1}\frac{c-x}{\sqrt{c^2+1-2cx}}\, dx.$$

$$= 4\pi - 2\pi c\int_{-1}^{1}\frac{dx}{\sqrt{c^2+1-2cx}} + 2\pi\int_{-1}^{1}\frac{x}{\sqrt{c^2+1-2cx}}\, dx$$

代入(i) $a = c^2+1,\ b = 2c.$

$$\int_{-1}^{1} \frac{dx}{\sqrt{c^2+1-2cx}} = \frac{2}{2c}\cdot\left(\sqrt{(c+1)^2} - \underbrace{\sqrt{(c-1)^2}}_{=c-1\ (c>1)}\right) = \frac{2}{c}.$$

代入(ii) $a = c^2+1,\ b = 2c.$

$$\int_{-1}^{1} \frac{x\,dx}{\sqrt{c^2+1-2cx}} = \frac{2}{4c^2}(c^2+1)\left(\sqrt{(c+1)^2} - \sqrt{(c-1)^2}\right)$$

$$- \frac{2}{3\times 4c^2}\left((c+1)^{\frac{3}{2}} - (c-1)^{\frac{3}{2}}\right).$$

$$= \frac{c^2+1}{2c^2}\times 2 - \frac{1}{6c^2}\left(6c^2+2\right).$$

$$= 1 + \frac{1}{c^2} - 1 - \frac{1}{3c^2} = \frac{2}{3c^2}.$$

$\Rightarrow$ 原式 $= 4\pi - 2\pi c\cdot\frac{2}{c} + 2\pi\cdot\frac{2}{3c^2} = \frac{4\pi}{3c^2}$

Rmk. 物理意义. 匀质球体对球外一点之万有引力可以看作将其质量集中于质心(球心)后，球心对那一点之万有引力.

变式. 如果 $0 < c \le 1$. 以上会发生什么变化?